PRACTICAL NO 3

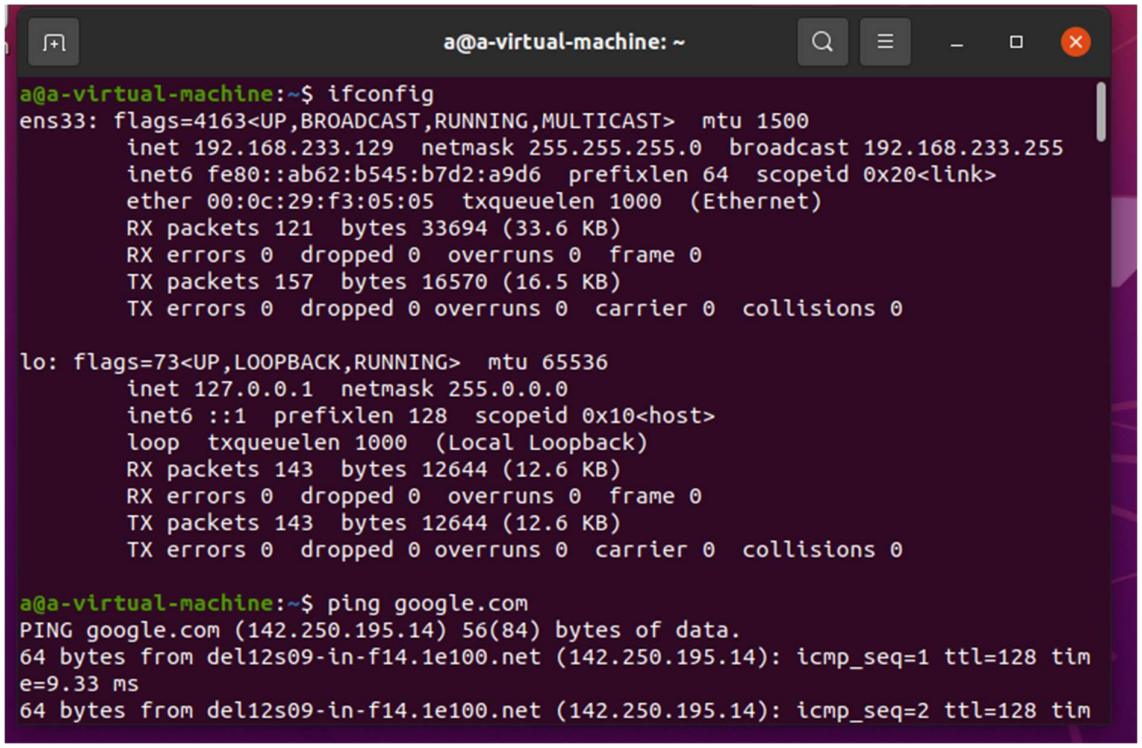
Aim: Linux Network Analysis and ARP Poisoning

● **Linux Network Analysis:**

Steps:

o Execute the ifconfig command to retrieve network interface information.

1.ipconfig



```
a@a-virtual-machine: ~
a@a-virtual-machine:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.233.129  netmask 255.255.255.0  broadcast 192.168.233.255
        inet6 fe80::ab62:b545:b7d2:a9d6  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:f3:05:05  txqueuelen 1000  (Ethernet)
        RX packets 121  bytes 33694 (33.6 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 157  bytes 16570 (16.5 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 143  bytes 12644 (12.6 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 143  bytes 12644 (12.6 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

a@a-virtual-machine:~$ ping google.com
PING google.com (142.250.195.14) 56(84) bytes of data.
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=1 ttl=128 tim
e=9.33 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=2 ttl=128 tim
```

o Use the ping command to test network connectivity and analyze the output.

2.ping google.com

```
a@a-virtual-machine: ~
a@a-virtual-machine:~$ ping google.com
PING google.com (142.250.195.14) 56(84) bytes of data.
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=1 ttl=128 tim
e=9.33 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=2 ttl=128 tim
e=6.15 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=3 ttl=128 tim
e=6.37 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=4 ttl=128 tim
e=5.63 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=5 ttl=128 tim
e=9.18 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=6 ttl=128 tim
e=7.42 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=7 ttl=128 tim
e=6.37 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=8 ttl=128 tim
e=7.68 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=9 ttl=128 tim
e=6.89 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=10 ttl=128 ti
me=6.67 ms
64 bytes from del12s09-in-f14.1e100.net (142.250.195.14): icmp_seq=11 ttl=128 ti
me=5.99 ms
```

o Analyze the netstat command output to view active network connections.

3.netstat

```
a@a-virtual-machine: ~
a@a-virtual-machine:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 a-virtual-machin:bootpc 192.168.233.254:bootps  ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         DGRAM                    49462    /run/systemd/notify
unix  2      [ ]         DGRAM                    108957   /run/user/1000/system
d/notify
unix  2      [ ]         DGRAM                    49476    /run/systemd/journal/
syslog
unix  18     [ ]         DGRAM                    49486    /run/systemd/journal/
dev-log
unix  8      [ ]         DGRAM                    49490    /run/systemd/journal/
socket
unix  3      [ ]         STREAM     CONNECTED     113436
unix  3      [ ]         STREAM     CONNECTED     111411   /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     100338
unix  3      [ ]         STREAM     CONNECTED     116088
unix  3      [ ]         STREAM     CONNECTED     113862   /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     112565
unix  3      [ ]         STREAM     CONNECTED     102967
unix  3      [ ]         STREAM     CONNECTED     114043
unix  3      [ ]         STREAM     CONNECTED     94909
```

o Perform a traceroute to trace the route packets take to reach a target host.

4.traceroute google.com

```
a@a-virtual-machine: ~
28  * *^C
a@a-virtual-machine:~$ sudo apt install traceroute
[sudo] password for a:
rSorry, try again.
[sudo] password for a:
Sorry, try again.
[sudo] password for a:
Reading package lists... Done
Building dependency tree
Reading state information... Done
traceroute is already the newest version (1:2.1.0-2).
0 upgraded, 0 newly installed, 0 to remove and 437 not upgraded.
1 not fully installed or removed.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Setting up linux-image-5.15.0-139-generic (5.15.0-139.149~20.04.1) ...
Processing triggers for linux-image-5.15.0-139-generic (5.15.0-139.149~20.04.1)
...
/etc/kernel/postinst.d/initramfs-tools:
update-initramfs: Generating /boot/initrd.img-5.15.0-139-generic
I: The initramfs will attempt to resume from /dev/sda5
I: (UUID=4f9ef8e8-add1-42ee-81bd-560179d80d0a)
I: Set the RESUME variable to override this.
```

```
a@a-virtual-machine: ~
a@a-virtual-machine:~$ traceroute google.com
traceroute to google.com (142.250.207.174), 30 hops max, 60 byte packets
 1  _gateway (192.168.233.2)  0.289 ms  0.191 ms  0.119 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
```

Windows:

1.ipconfig

```
Command Prompt
C:\Users\ITCS>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6b22:cdb:26ca:74d9%16
   IPv4 Address. . . . . . . . . . . : 192.168.105.59
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.105.11

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::468d:1b69:a1d4:1178%10
   IPv4 Address. . . . . . . . . . . : 192.168.221.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::c291:6169:5b64:5f48%14
   IPv4 Address. . . . . . . . . . . : 192.168.233.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

C:\Users\ITCS>
```

2.ping

```
Command Prompt
C:\Users\ITCS>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
```

3.netstat

```
Command Prompt
C:\Users\ITCS>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49671        COMPedu8024:49672      ESTABLISHED
  TCP    127.0.0.1:49672        COMPedu8024:49671      ESTABLISHED
  TCP    127.0.0.1:49673        COMPedu8024:49674      ESTABLISHED
  TCP    127.0.0.1:49674        COMPedu8024:49673      ESTABLISHED
  TCP    127.0.0.1:54353        COMPedu8024:54354      ESTABLISHED
  TCP    127.0.0.1:54354        COMPedu8024:54353      ESTABLISHED
  TCP    127.0.0.1:54396        COMPedu8024:54397      ESTABLISHED
  TCP    127.0.0.1:54397        COMPedu8024:54396      ESTABLISHED
  TCP    192.168.105.59:54290   13.107.5.93:https      ESTABLISHED
  TCP    192.168.105.59:54293   sc-in-f188:5228        ESTABLISHED
  TCP    192.168.105.59:54337   4.213.25.240:https     ESTABLISHED
  TCP    192.168.105.59:54405   a23-217-111-112:https  LAST_ACK
  TCP    192.168.105.59:54406   a23-217-111-112:https  LAST_ACK
  TCP    192.168.105.59:54407   a23-217-111-112:https  LAST_ACK
  TCP    192.168.105.59:54409   a23-217-111-112:https  ESTABLISHED
  TCP    192.168.105.59:54412   49.44.217.32:https     LAST_ACK
  TCP    192.168.105.59:54453   146.75.118.172:http    ESTABLISHED
  TCP    192.168.105.59:54454   bom12s14-in-f14:https  ESTABLISHED
  TCP    192.168.105.59:54456   bom12s03-in-f10:https  ESTABLISHED
  TCP    192.168.105.59:54457   bom12s03-in-f10:https  ESTABLISHED
  TCP    192.168.105.59:54460   49.44.217.32:https     ESTABLISHED
  TCP    192.168.105.59:54461   204.79.197.222:https   ESTABLISHED
  TCP    192.168.105.59:54462   131.253.33.254:https   ESTABLISHED
  TCP    192.168.105.59:54463   20.141.12.34:https     ESTABLISHED
  TCP    [::1]:54398            COMPedu8024:54399      ESTABLISHED
```

4.tracert www.google.com

```
Command Prompt - tracert  www.google.com
  TCP    [::1]:54399            COMPedu8024:54398      ESTABLISHED

C:\Users\ITCS>tracert www.google.com

Tracing route to www.google.com [142.251.156.119]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.105.11
  2     4 ms     1 ms     1 ms  reliance.reliance [192.168.29.1]
  3     6 ms     3 ms     3 ms  10.228.160.1
  4     6 ms     4 ms     6 ms  172.31.2.22
  5     3 ms     4 ms     3 ms  192.168.70.16
  6
```